

Azure Full Configuration Reference

Every service, API, database, and setting required to run AtlasRide on Azure App Services — with specific values where known.

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This document contains environment variable names and service configuration details. Keep it secure. Never commit real secrets to version control.

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1. Azure App Service Settings

Configuration !' General Settings in the Azure Portal

Setting Name	Value / Description	Required
Runtime Stack	Node.js	Required
Node.js Version	24 LTS (matches GitHub Actions workflow)	Required
Startup Command	npm run start	Required
Platform	Linux	Required
Always On	ON — prevents sleep and WebSocket disconnections	Required
WebSockets	ON — required for real-time driver tracking and in-ride chat	Required
ARR Affinity	ON — keeps sessions on the same instance (single instance)	Recommended
HTTPS Only	ON — required for secure cookies to work	Required
Minimum TLS Version	1.2	Required

The startup command "npm run start" runs: `NODE_ENV=production node dist/index.cjs` — the pre-compiled server bundle built by GitHub Actions.

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2. PostgreSQL Database

Azure Database for PostgreSQL — Flexible Server (recommended)

2a. Create the Database Server

In Azure Portal !' Create a resource !' Azure Database for PostgreSQL Flexible Server

Setting Name	Value / Description	Required
Server type	Flexible Server (recommended over Single Server)	Required
PostgreSQL version	16 (or 15 — both work)	Required
Compute tier	Burstable B1ms (1 vCore, 2 GB) is enough to start	Recommended
Admin username	Choose a username (e.g. atlasride_admin)	Required
Admin password	Strong password — 8+ chars, upper, lower, number, symbol	Required
Allow Azure services	YES — needed so the App Service can connect	Required
SSL enforcement	Enabled (default)	Required

2b. Connection String Format

```
DATABASE_URL format — paste this into Azure App Settings
postgres://USERNAME:PASSWORD@SERVER.postgres.database.azure.com/DATABASE_NAME?sslmode=require

# Example:
postgres://atlasride_admin:MyPass123!@atlasride-db.postgres.database.azure.com/atlasride?sslmode=require
```

The `?sslmode=require` suffix is mandatory. Without it Azure PostgreSQL refuses the connection.

2c. Run Schema Migration (one-time setup)

Run this once from your local machine after the Azure database is created. Point DATABASE_URL at your Azure database temporarily:

Terminal — run from project root

```
DATABASE_URL="postgresql://user:pass@host/db?sslmode=require" npm run db:push
```

This creates all tables including the sessions table which is required for logins to work.

Ø=Ý 3. Environment Variables — Complete Reference

Azure Portal ! App Service ! Configuration ! Application Settings ! + New application setting

Set ALL of these in Azure App Settings. Do NOT set them in .env files — those are for local development only. Variables marked AUTO are set by Azure automatically; never override them.

Setting Name	Value / Description	Required
NODE_ENV	production	REQUIRED
DATABASE_URL	postgresql://user:pass@host/db?sslmode=require	REQUIRED
SESSION_SECRET	Random 40+ character string — generate with: openssl rand -base64 40	REQUIRED
AZURE_MAPS_KEY	Your Azure Maps subscription key (see section 4)	REQUIRED
STRIPE_SECRET_KEY	sk_live_... or sk_test_... (from Stripe Dashboard)	REQUIRED
STRIPE_PUBLISHABLE_KEY	pk_live_... or pk_test_... (from Stripe Dashboard)	REQUIRED
TWILIO_ACCOUNT_SID	ACxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx (from Twilio Console)	REQUIRED
TWILIO_AUTH_TOKEN	Your Twilio Auth Token (from Twilio Console)	REQUIRED
TWILIO_MESSAGING_SERVICE_SID	MGxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx (from Twilio Console)	REQUIRED
ENTRA_CLIENT_ID	Application (client) ID from Azure Entra app registration (see section 7)	REQUIRED
ENTRA_CLIENT_SECRET	Client secret value from Azure Entra app registration (see section 7)	REQUIRED
MAPBOX_ACCESS_TOKEN	pk.eyJ1... — Mapbox public token (optional, see section 8)	Optional
PORT	8080 — SET BY AZURE AUTOMATICALLY. Do not override.	AUTO
WEBSITE_HOSTNAME	saviaj.azurewebsites.net — SET BY AZURE AUTOMATICALLY.	AUTO

Generate a SESSION_SECRET

Run any one of these to produce a secure random secret

Option 1 — Mac / Linux terminal:
openssl rand -base64 40

Option 2 — PowerShell (Windows):
[Convert]::ToBase64String((1..40 | ForEach-Object { Get-Random -Maximum 256 })))

Option 3 — Node.js:
node -e "console.log(require('crypto').randomBytes(40).toString('base64'))"

4. Azure Maps

Geocoding, reverse geocoding, and route distance calculation

What AtlasRide uses Azure Maps for:

Setting Name	Value / Description
Forward geocoding	atlas.microsoft.com/search/address/json — converts typed address to lat/lng
Reverse geocoding	atlas.microsoft.com/search/address/reverse/json — converts lat/lng to readable address
Route directions	atlas.microsoft.com/route/directions/json — calculates driving distance between pickup and dropoff

4a. Create an Azure Maps Account

1

Azure Portal !' Create a resource !' search "Azure Maps"

Select Azure Maps !' Create
Choose your subscription and resource group
Name: e.g. atlasride-maps
Pricing tier: Gen2 (S0 is fine for production — pay-per-use)

2

Get your subscription key

Azure Portal !' your Azure Maps account !' Settings !' Authentication
Copy "Primary Key" — this is your AZURE_MAPS_KEY value
Paste it into Azure App Settings as AZURE_MAPS_KEY

Azure Maps Gen2 pricing: geocoding is ~\$0.50 per 1,000 requests. Route directions ~\$0.50 per 1,000 requests. Very low cost at typical ride volumes.

4b. Known Values (already in the code — no action needed)

Setting Name	Value / Description
API base URL	https://atlas.microsoft.com
Auth method	Subscription key (via AZURE_MAPS_KEY header)
API version	2022-08-01 (set automatically by SDK)

5. Stripe Payments

Payment processing for ride bookings

What AtlasRide uses Stripe for:

Riders pay for booked rides via Stripe Checkout. Stripe webhooks notify the server when payment is confirmed, triggering ride status updates. Driver earnings are tracked and protected server-side.

5a. Get your Stripe API Keys

1

Log in to your Stripe Dashboard !' Developers !' API keys

Copy "Publishable key" (starts with pk_live_ or pk_test_)
Copy "Secret key" (starts with sk_live_ or sk_test_)
Add both to Azure App Settings (see section 3)

2

Switch to Live Mode for production

In Stripe Dashboard, toggle from "Test mode" to "Live mode" (top-left toggle)
The live keys start with pk_live_ and sk_live_

Use test keys (pk_test_, sk_test_) during testing only

5b. Stripe Webhook (auto-configured)

The webhook URL and secret are managed automatically by the app. After first deployment, verify in Stripe Dashboard:

```
Stripe Dashboard !' Developers !' Webhooks
# Webhook URL format (auto-built from WEBSITE_HOSTNAME):
https://YOUR-DOMAIN.azurewebsites.net/api/stripe/webhook/{uuid}

# Verify in Stripe Dashboard !' Developers !' Webhooks
# Status should show as "Enabled" with events: * (all events)
```

Setting Name	Value / Description
Stripe API version	2025-11-17.clover (hardcoded in stripeClient.ts)
Webhook events	* (all events — managed automatically)
Webhook secret	Auto-managed by stripe-replit-sync library — no manual action needed

Stripe is optional for the app to START. If STRIPE_SECRET_KEY and STRIPE_PUBLISHABLE_KEY are missing, the app still launches but payment routes return errors. Add them before taking your first real booking.

6. Twilio SMS

Phone number verification during user registration

What AtlasRide uses Twilio for:

When a user registers or changes their phone number, a 6-digit OTP is sent via SMS to verify ownership. Twilio Messaging Services are used (not a direct phone number).

6a. Setup Steps

- 1

Create a Twilio account at twilio.com

Complete account verification

From the Twilio Console Dashboard, copy:

- Account SID (starts with AC...) ! TWILIO_ACCOUNT_SID
 - Auth Token ! TWILIO_AUTH_TOKEN
- 2

Create a Messaging Service

Twilio Console ! Messaging ! Services ! Create Messaging Service

Name: e.g. AtlasRide SMS

Purpose: Notifications

Add a phone number (buy one in Phone Numbers ! Manage ! Buy a number)

After creating, copy the Messaging Service SID (starts with MG...) ! TWILIO_MESSAGING_SERVICE_SID
- 3

Enable geographic permissions

Twilio Console ! Messaging ! Settings ! Geo permissions

Enable the countries your users are in (UK, etc.)

Without this, international SMS will fail silently

Setting Name	Value / Description
TWILIO_ACCOUNT_SID	Found in Twilio Console ! Dashboard (top of page)
TWILIO_AUTH_TOKEN	Found in Twilio Console ! Dashboard ! click to reveal
TWILIO_MESSAGING_SERVICE_SID	Found in Twilio Console ! Messaging ! Services ! your service ! Properties
Phone number format	E.164 international format enforced (e.g. +447911123456)

Twilio SMS is required for phone OTP during registration. If Twilio is not configured the app still starts, but phone verification will fail and users cannot complete registration.

7. Microsoft Entra External ID (Email OTP)

Email verification for signup and password reset — tenant: atlasridecustomers

What AtlasRide uses Entra for:

When a user registers or resets their password, a 6-digit OTP is sent to their email by Microsoft's own email delivery infrastructure. No email provider to configure separately.

7a. Your Tenant Details (already configured in the code)

Setting Name	Value / Description
Tenant name	atlasridecustomers (already hardcoded in entraEmailOtp.ts)
Tenant domain	atlasridecustomers.onmicrosoft.com (already set)
CIAM endpoint	https://atlasridecustomers.ciamlogin.com (already set)

7b. Get your Client ID and Secret

1

Sign in to portal.azure.com !' Microsoft Entra ID

In the left menu: App registrations

Find your app registration (or create a new one for AtlasRide)

Copy the "Application (client) ID" !' `ENTRA_CLIENT_ID`

2

Create a client secret

In your app registration !' Certificates & secrets !' + New client secret

Description: AtlasRide Azure Production

Expiry: 24 months (set a calendar reminder to rotate it)

After clicking Add, copy the "Value" immediately !' `ENTRA_CLIENT_SECRET`

IMPORTANT: The value is only shown once. Copy it now.

3

Ensure Native Auth is enabled

In your app registration !' Authentication !' Advanced settings

Enable "Allow public client flows" !' Yes

In the CIAM admin center, ensure "Enable Native Authentication" is ON for this tenant

`ENTRA_CLIENT_SECRET` expires. Set a calendar reminder before the expiry date to generate a new secret and update the Azure App Setting — otherwise email OTP will stop working silently.

8. Mapbox (Optional)

Map tile rendering — currently handled by OpenStreetMap by default

Mapbox is referenced in the codebase but is currently optional. The map rendering uses OpenStreetMap tiles by default (free, no key needed). If you switch to Mapbox tiles for a premium look, you need this token.

1

- Create a Mapbox account at mapbox.com
- Go to Account !' Access tokens
- Copy the default public token (starts with pk.eyJ1...)
- Add to Azure App Settings as MAPBOX_ACCESS_TOKEN

You do NOT need Mapbox to run AtlasRide. The app works fully with OpenStreetMap tiles. Add this only if you decide to switch to Mapbox for map rendering.

9. GitHub Actions Deployment Workflow

Your workflow file: `.github/workflows/main_saviaj.yml` — already in the repo

How deployment works:

Every push to the main branch triggers the workflow automatically. It:

Setting Name	Value / Description
Step 1	Checks out code from GitHub
Step 2	Installs Node 24, runs npm install (all dependencies)
Step 3	Runs npm run build !' produces dist/index.cjs + dist/public/
Step 4	Uploads the full directory (with dist/ included) as artifact
Step 5	Logs into Azure using OIDC (no publish profile needed)
Step 6	Deploys the artifact to Azure App Service "Saviaj", Production slot

GitHub Secrets already wired (set by Azure Deployment Center):

Setting Name	Value / Description
AZUREAPPSERVICE_CLIENTID_720A3FC2...	Azure service principal client ID — already set in GitHub repo secrets
AZUREAPPSERVICE_TENANTID_171C2A6F...	Azure tenant ID — already set in GitHub repo secrets
AZUREAPPSERVICE_SUBSCRIPTIONID_7A9...	Azure subscription ID — already set in GitHub repo secrets

These secrets were added automatically when you connected your GitHub repo via Azure Deployment Center. No manual action needed for deployment auth.

To deploy a new version:

- Terminal — run from project root
- # 1. Commit your changes
- git add -A
- git commit -m "your message"
- # 2. Push to main — this triggers the GitHub Actions workflow automatically
- git push
- # 3. Monitor deployment
- # GitHub !' your repo !' Actions tab !' watch the workflow run (~3-5 minutes)

Check deployment status:

GitHub !' your repository !' Actions tab. A green tick means the deployment succeeded. A red cross means something failed — click the run to see which step failed and the error message.

10. Go-Live Checklist

Tick everything before opening AtlasRide to real users

Azure App Service Configuration

- ☐ **Runtime: Node.js 24 LTS**
Configuration ! General Settings ! Stack
- ☐ **HTTPS Only: ON**
Configuration ! General Settings
- ☐ **WebSockets: ON**
Configuration ! General Settings
- ☐ **Startup: npm run start**
Configuration ! General Settings ! Startup Command
- ☐ **Always On: ON**
Configuration ! General Settings
- ☐ **ARR Affinity: ON**
Configuration ! General Settings

Environment Variables (Azure App Settings)

- ☐ **NODE_ENV = production**
Application Settings
- ☐ **SESSION_SECRET = 40+ random chars**
Application Settings
- ☐ **STRIPE_SECRET_KEY set**
Application Settings
- ☐ **TWILIO_ACCOUNT_SID set**
Application Settings
- ☐ **TWILIO_MESSAGING_SERVICE_SID set**
Application Settings
- ☐ **ENTRA_CLIENT_SECRET set**
Application Settings
- ☐ **DATABASE_URL set with ?sslmode=require**
Application Settings
- ☐ **AZURE_MAPS_KEY set**
Application Settings
- ☐ **STRIPE_PUBLISHABLE_KEY set**
Application Settings
- ☐ **TWILIO_AUTH_TOKEN set**
Application Settings
- ☐ **ENTRA_CLIENT_ID set**
Application Settings

Database

- ☐ **Azure PostgreSQL server created**
Flexible Server, SSL enabled
- ☐ **"Allow Azure services" is ON**
Networking ! Firewall rules
- ☐ **Database created (e.g. atlasride)**
Azure Portal ! your server ! Databases
- ☐ **npm run db:push run against Azure DB**
Creates all tables incl. sessions

Stripe

- ☐ **Live mode keys set (pk_live_, sk_live_)**
Not test keys for production
- ☐ **Test payment completed successfully**
Use Stripe test card 4242 4242 4242 4242
- ☐ **Webhook visible in Stripe Dashboard**
Developers ! Webhooks ! status: Enabled

Twilio

- ☐ **Messaging Service created**
With at least one phone number attached
- ☐ **Test SMS received on a real phone**
Register a test account, verify number
- ☐ **Geo permissions enabled for UK/target countries**
Messaging ! Settings ! Geo permissions

Entra External ID

- ☐ **App registration found/created**
portal.azure.com ! Entra ID ! App registrations
- ☐ **Client secret created and copied**
ENTRA_CLIENT_SECRET — copy immediately after creation
- ☐ **ENTRA_CLIENT_ID copied**
Application (client) ID
- ☐ **Native Authentication enabled**
Allow public client flows: Yes

Secret expiry noted in calendar

Rotate before it expires or email OTP breaks

Post-Deployment Verification

- ☐ **Home page loads at your .azurewebsites.net URL**
Should show AtlasRide login screen
- ☐ **Register a new account**
Tests DB, Entra email OTP, and sessions
- ☐ **Post a ride request**
Tests Azure Maps geocoding
- ☐ **Log stream shows no ERROR lines**
Azure Portal !' App Service !' Log stream
- ☐ **/api/auth/user returns 401**
Confirms API is live (normal before login)
- ☐ **Verify phone number via SMS**
Tests Twilio integration end-to-end
- ☐ **Complete a test payment**
Tests Stripe end-to-end (use test card)

After passing all checks above, your AtlasRide instance is ready for real users. Keep this document updated whenever you rotate secrets or add new services.

